

Business Data Analytics Project Report

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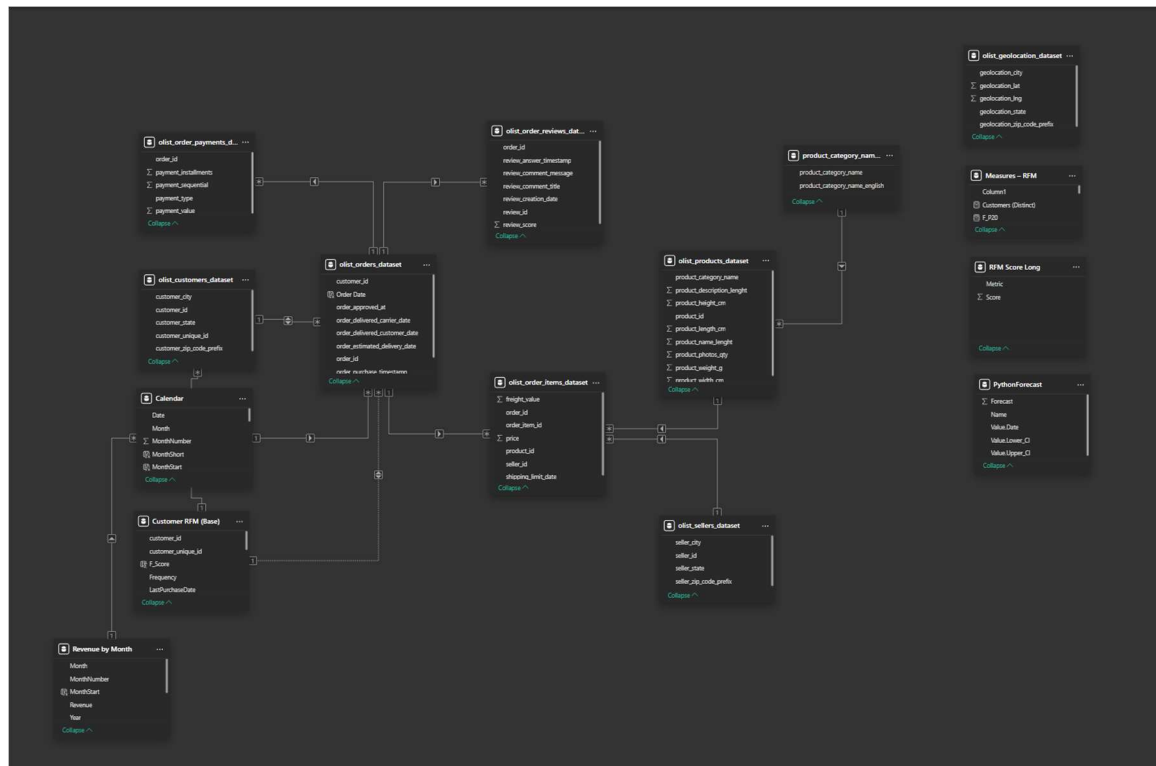
1. Introduction

This report presents a complete Business Data Analytics project built using the Olist E-commerce dataset. It includes data modelling, DAX calculations, customer segmentation, revenue analysis, and Python-based forecasting using a SARIMAX time-series model. All visuals are generated in Power BI, and the forecasting model was implemented externally in Python and integrated back into the dashboard.

2. Project Assumptions

- The dataset represents real e-commerce transactions from Olist Brazil for the years 2016–2018.
- The goal is to build dashboards for executives, marketing teams, and financial analysts.
- Python forecasting is executed using monthly aggregated revenue.
- RFM segmentation uses Recency, Frequency, and Monetary values derived from customer purchase history.

3. Data Model



The data model represents the core backbone of the entire analytical system. It consolidates multiple raw Olist datasets into a clean, relational, analytics-ready structure.

A star-schema-inspired hybrid design was used to optimize performance, enable easy DAX modeling, and ensure accurate filtering across dashboards.

◆ 1. Overall Modeling Approach

The model combines multiple Olist tables:

- Orders
- Customers
- Sellers
- Products
- Geolocation
- Order Items
- Order Payments
- Order Reviews

These were structured around a central fact table (olist_order_items_dataset) with several dimension tables feeding into it.

In addition, custom analytical tables were created:

- **Calendar table** (for date intelligence)
- **Customer RFM table**
- **Revenue by Month table**
- **Python Forecast table**

This hybrid approach enables:

- ✓ advanced time-series analysis
- ✓ RFM segmentation
- ✓ category/seasonality analysis
- ✓ forecasting integration

The model's relationships were carefully designed to ensure directional filtering behaves correctly across dashboards.

◆ 2. Central Fact Table

Fact Table: olist_order_items_dataset

This table acts as the main fact table and contains:

- order_id
- order_item_id
- product_id
- seller_id
- price
- freight_value
- shipping_limit_date

It connects sales revenue with product, seller, and order information.

This is the table from which most quantitative measures (Total Sales, Revenue, Quantity, AOV) originate.

◆ 3. Dimension Tables

a) olist_orders_dataset

Provides order-level metadata:

- order_purchase_timestamp
- order_approved_at
- order_delivered_carrier_date
- order_delivered_customer_date
- order_estimated_delivery_date

Relationship:

- Linked to the fact table via order_id
 - Filters all fact data chronologically through the Calendar table.
-

b) olist_customers_dataset

Contains unique customer information:

- customer_unique_id
- customer_city
- customer_state
- customer_zip_code_prefix

Relationship:

- Linked to olist_orders_dataset via customer_id
 - Enables segmentation by region, customer attributes, and RFM grouping.
-

c) olist_products_dataset

Provides product-level metadata:

- product_category_name
- product_dimensions
- product_weight
- product_description_length

Relationship:

- Linked to olist_order_items_dataset via product_id
 - Used for category-based insights, mix analysis, and seasonality trends.
-

d) olist_sellers_dataset

Includes seller information (ID, city, state, ZIP code prefix).

Relationship:

- Linked to fact table via seller_id
 - Enables regional seller performance analytics.
-

e) olist_geolocation_dataset

Contains raw geographic coordinates.

Although not used directly in visuals, it supports advanced geographical enrichment if required later.

f) product_category_name_translation

Maps Portuguese category names into English.

This ensures:

- Clear category labels
 - Consistency across dashboards
 - Improved interpretability for non-Portuguese readers
-

◆ 4. Custom Analytical Tables

a) Calendar Table (Date Dimension)

The Calendar table is fully independent and includes:

- Date
- Month
- MonthNumber
- MonthShort
- Year

This table is essential for:

- ✓ time intelligence functions
 - ✓ forecasting alignment
 - ✓ seasonality/heatmap visuals
 - ✓ year-over-year comparisons
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b) Customer RFM Table (RFM Base)

This table contains pre-computed:

- Recency
- Frequency
- Monetary
- RFM Score

It merges customer behavior insights with transactional data, powering the Customer Segmentation (RFM) dashboard.

It joins into:

- the Calendar table for recency calculations
 - the Customer table via customer_unique_id
-

c) Revenue by Month Table

This aggregated table stores:

- Month
- MonthNumber
- Revenue
- Year

Purpose:

- Simplifies forecasting input
 - Ensures a clean, chronological time series
 - Feeds monthly revenue visuals across multiple dashboards
-

d) Python Forecast Table

Created by exporting Python output:

- Forecast value
- Lower bound (ValueLower_CI)

- Upper bound (ValueUpper_CI)
- Date

This table powers:

- ✓ the forecasting KPI cards
 - ✓ the integration of Python SARIMAX predictions
 - ✓ the chart filters and slicers affecting Python results
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◆ 5. Relationship Structure

The model uses single-direction relationships where possible to prevent ambiguity and maintain performance.

Few relationship highlights:

✓ Orders → Order Items

(1:* relationship)

Ensures each order is linked to its item-level transactions.

✓ Products → Order Items

(1:* relationship)

Enables category-based filtering.

✓ Customers → Orders

(1:* relationship)

Links customer-level data to purchase behavior.

✓ Calendar → Orders

(1:* relationship)

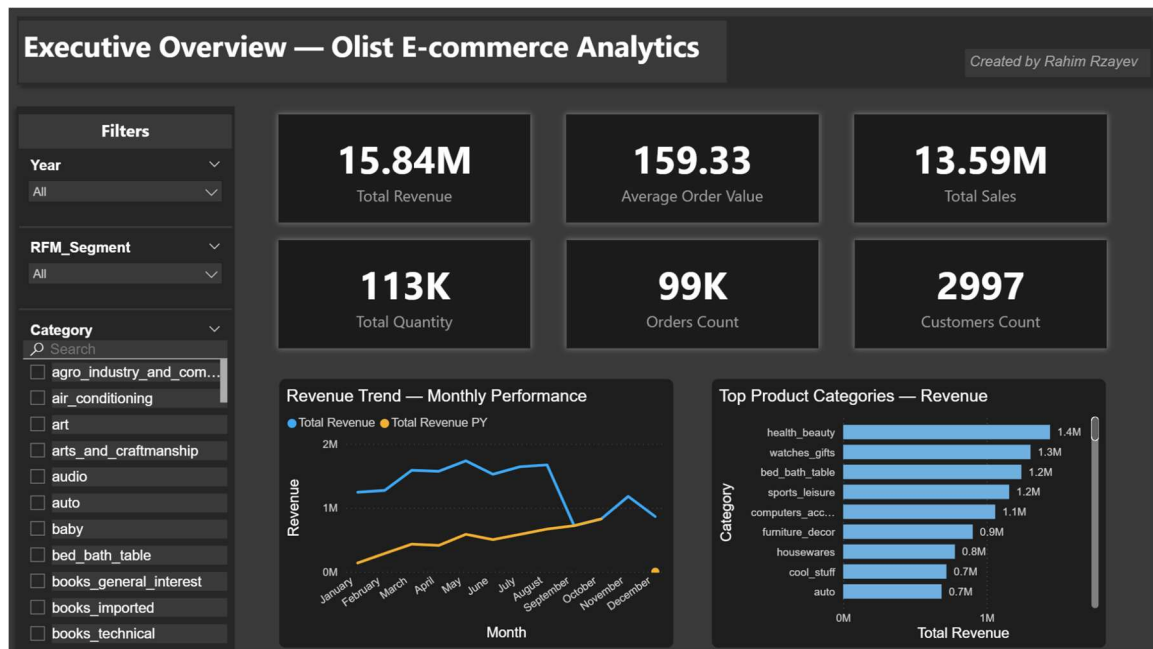
Allows proper filtering of all time-dependent visuals.

✓ RFM Table → Customers

(1:* relationship)

Supports segmentation and behavioral clustering.

4. Executive Overview Dashboard



The **Executive Overview** dashboard serves as the high-level summary page of the entire BI solution. It is designed for decision-makers who need a quick, precise understanding of the business's overall performance without navigating multiple pages. All critical KPIs, trends, and category-level breakdowns are displayed in a clear and intuitive layout.

◆ 1. Filters Panel (Year, RFM Segment, Category)

This section allows users to dynamically explore the data:

Year Filter

- Enables time-based slicing for annual analysis.
- Supports year-over-year comparisons when interacting with trend charts and KPIs.

RFM_Segment Filter

- Allows filtering revenue and customer behavior by recency, frequency, and monetary segmentation.
- Provides insight into how Loyalists, Lost Customers, and Potential Loyalists contribute to revenue.

Category Filter (+ Search)

- Allows selecting individual product groups.

- Includes a search bar to quickly locate categories among 70+ product types.
- Automatically updates all visuals, enabling deep-dive category-level exploration.

These filters create a fully dynamic dashboard where managers can quickly isolate patterns across time periods, customer groups, and product lines.

◆ 2. KPI Cards — Business Performance Snapshot

The top section includes six KPIs:

✓ Total Revenue (15.84M)

Sum of all order item sales.

Shows overall business performance and scale.

✓ Average Order Value (159.33)

Formula:

$$\text{AOV} = \frac{\text{Total Revenue}}{\text{Number of Orders}}$$

Measures customer spending behavior.

✓ Total Sales (13.59M)

Represents total GMV from completed transactions.

Highlights difference between sales vs. revenue (e.g., returns, cancellations).

✓ Total Quantity (113K)

Indicates total items sold.

Identifies product demand volume beyond monetary value.

✓ Orders Count (99K)

Shows total number of customer transactions.

Useful for operational insights and logistics planning.

✓ Customers Count (2997)

Unique customer count using DISTINCT.

Key indicator of market size and retention challenge.

This KPI layer gives executives instant visibility into revenue health, customer base, and average spending—critical for high-level decision-making.

◆ 3. Revenue Trend — Monthly Performance (Line Chart)

This chart compares:

- **Total Revenue (current year)**
- **Total Revenue PY (previous year)**

Key insights enabled:

- Month-over-month revenue seasonality
- Identification of growth spikes or dips
- YOY performance tracking
- Detecting high-demand seasonal months (e.g., holiday season)

Technical Elements Used:

- `CALCULATE()` with `SAMEPERIODLASTYEAR()` DAX
- Calendar table relationship
- Line chart with dual-series comparison

This chart adds strong analytical depth because it highlights growth patterns and reveals operational bottlenecks or opportunities.

◆ 4. Top Product Categories — Revenue (Bar Chart)

This visual shows a ranked bar chart of categories contributing most to overall revenue.

Top performers identified:

- health_beauty
- watches_gifts
- bed_bath_table
- sports_leisure
- computers_accessories

Why this is important:

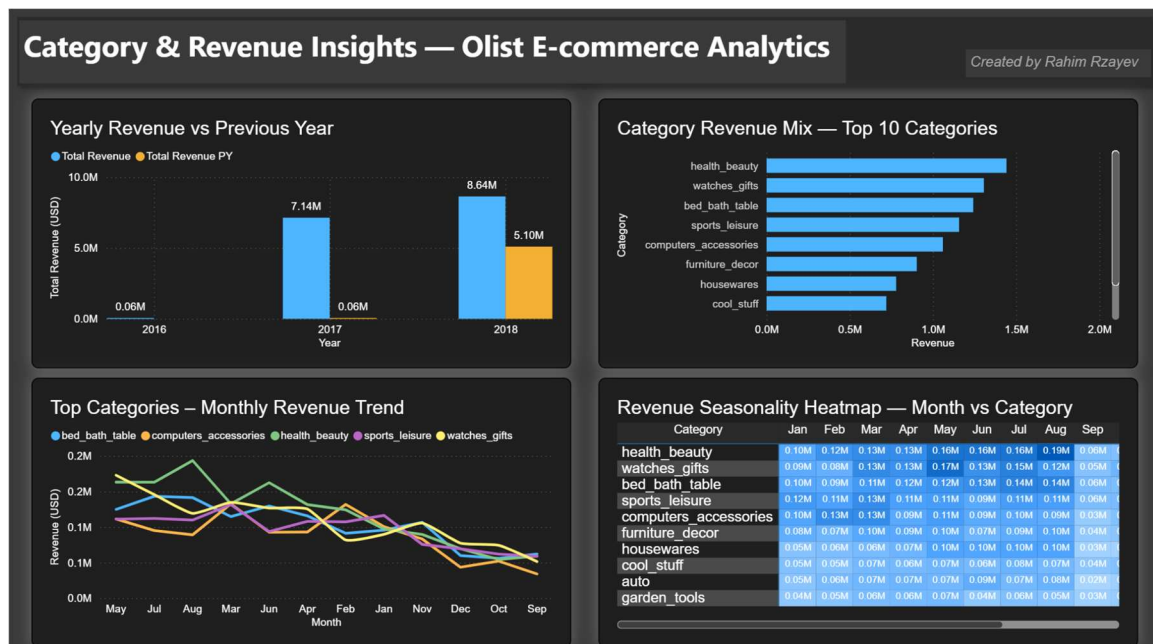
- Helps prioritize inventory decisions
- Supports marketing spend allocation
- Allows identifying weak categories dragging overall revenue

Technical Work:

- Aggregation of revenue by category using SUMX
- Sorting by descending revenue
- Scrollbar due to large category count (over 70+)

This chart gives management clarity on which categories are driving financial performance.

5. Category & Revenue Insights Dashboard



The **Category & Revenue Insights** dashboard provides a deeper analytical breakdown of how individual product categories contribute to overall revenue performance over time. Unlike the Executive Overview page, which focuses on high-level KPIs, this dashboard explores category-level behavior, seasonality, and multi-year comparative trends, offering a more granular understanding of product demand patterns.

◆ 1. Yearly Revenue vs Previous Year (Bar Chart)

Purpose

This chart compares **Total Revenue** with **Total Revenue PY** (previous year) across multiple years. It highlights year-over-year growth and structural changes in sales performance.

What We Did Technically

- Created DAX measures using **SAMEPERIODLASTYEAR()**
- Joined the Calendar table properly to ensure time intelligence works
- Used column labels & data labels for clarity
- Added contrasting colors (blue for current year, yellow for previous year)

Insights Provided

- The revenue growth in 2017 and 2018 is clearly visible, showing that the business experienced upward momentum.
- The difference between revenue and PY helps identify growth acceleration or slowdowns.
- The very low value in 2016 suggests that data availability starts mid-year, which is common in Olist datasets.

This visual is essential for understanding long-term trends and validating that the business is scaling sustainably.

◆ 2. Category Revenue Mix — Top 10 Categories (Horizontal Bar Chart)

Purpose

This visual highlights the **top 10 revenue-generating categories**, ranked in descending order.

How We Built It

- Created a revenue measure using SUMX() over order items
- Sorted categories by total revenue
- Applied a scroll bar for usability because the Olist dataset contains 70+ categories
- Used a horizontal layout to maximize readability for long category names

Key Business Insights

- health_beauty, watches_gifts, and bed_bath_table dominate the market.
- Revenue distribution is not uniform — a small number of categories capture the majority of revenue.
- The insight helps determine:
 - Which product groups deserve higher marketing budgets
 - Where to focus inventory and logistics efforts
 - Which categories may have hidden growth potential if revenue is low but trending upward

This ranked bar chart is one of the most valuable visuals for strategic category management.

◆ 3. Top Categories – Monthly Revenue Trend (Multi-Series Line Chart)

Purpose

This chart compares the monthly revenue patterns of the top-performing categories to uncover:

- Seasonal peaks
- Demand volatility
- Stability or inconsistency across months
- Competition between categories

Technical Construction

- Selected top 5–6 categories based on revenue
- Built separate DAX measures for each category
- Implemented a Calendar relationship for correct month sorting
- Applied a distinct color for each category line

Insights

- Categories like health_beauty and sports_leisure show strong seasonal spikes.
- Some categories have stable, constant revenue (e.g., computers_accessories).
- The visual helps detect:
 - Q4 peak periods
 - Low-demand months
 - Categories sensitive to holidays or back-to-school seasons

This chart is crucial for demand planning and forecasting, revealing month-to-month fluctuations.

◆ 4. Revenue Seasonality Heatmap — Month vs Category

Purpose

The heatmap provides an intuitive visualization of seasonality patterns across categories and months. Darker cells indicate higher revenue, while lighter cells reflect lower demand.

Technical Work Included

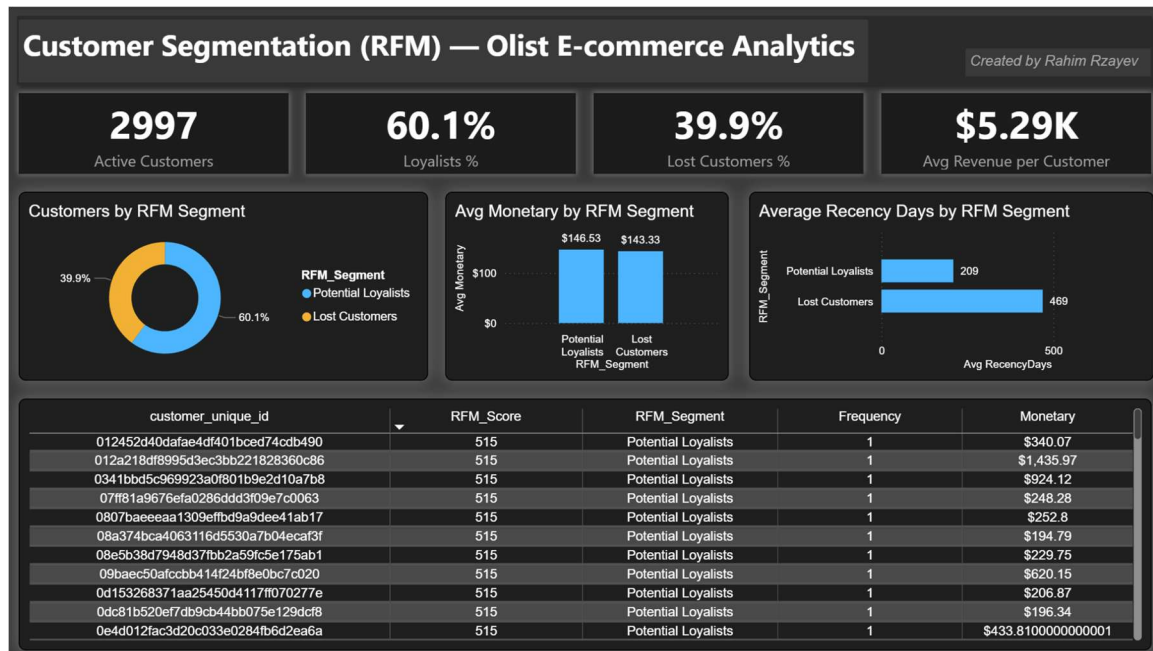
- Created a matrix visual
- Applied conditional formatting (background color scaling)
- Ensured correct month order using the MonthNumber column
- Formatted values in millions for clean readability

Insights Offered

- health_beauty and watches_gifts repeatedly peak from May to August, suggesting warm-season or promotional influences.
- Categories like auto and garden_tools show very stable but lower revenue patterns.
- Seasonality clusters become obvious:
 - Summer peaks
 - Late-year declines
 - Category-specific cycles

This heatmap is extremely powerful for operational functions such as inventory planning, warehouse staffing, marketing timing, and sales forecasting.

6. Customer Segmentation (RFM) Dashboard



The **Customer Segmentation (RFM)** dashboard applies the RFM analytical model to classify customers based on their transactional behavior. RFM (Recency, Frequency, Monetary) is a widely used customer analytics methodology in e-commerce and CRM, allowing organizations to evaluate customer value, retention, and loyalty patterns.

This dashboard translates customer activity in the Olist marketplace into clear, interpretable, and actionable customer segments.

◆ 1. Customer KPIs (Top-of-Dashboard Metrics)

Active Customers — 2,997

This metric shows the total number of unique customers present in the dataset. It is calculated using:

Customers = `DISTINCTCOUNT(olist_customers_dataset[customer_unique_id])`

Loyalists % — 60.1%

Percentage of customers classified as **Potential Loyalists** based on their RFM scores.

Lost Customers % — 39.9%

Percentage of customers showing long inactivity periods (high recency value).

Avg Revenue per Customer — \$5.29K

Computed by dividing total revenue by the number of active customers.

Insights

- More than half of the customer base demonstrates loyal behavior.
 - Despite a significant portion of lost customers, revenue concentration among active ones remains strong.
-

◆ 2. Donut Chart: Customers by RFM Segment

Purpose

This visual provides a quick distribution view of how customers are segmented based on their RFM scores.

Segments displayed:

- **Potential Loyalists (60.1%)**
- **Lost Customers (39.9%)**

How It Was Built

- We calculated an RFM score for each customer using weighted recency, frequency, and monetary metrics.
- Customers were classified using conditional DAX logic (e.g., SWITCH() or nested IF statements).
- A donut chart was selected to make the segment proportions visually intuitive.

Key Insight

The business has a healthy core of loyal customers, but a meaningful share of the base is at risk of churn.

◆ 3. Avg Monetary by RFM Segment (Bar Chart)

Purpose

This chart helps identify how much money each customer segment spends on average.

Technical Steps

- Created a Monetary measure summarizing total spending per customer.
- Aggregated and averaged Monetary values within each segment.
- Used neutral color bars to maintain report consistency.

Business Insight

- Potential Loyalists and Lost Customers have similar average order value (~\$145).
- The similarity indicates that monetary value does not differ dramatically; segmentation is driven more by behavior (recency & frequency) than spending power.

This helps determine whether to push reactivation campaigns or focus on nurturing loyal buyers.

◆ 4. Average Recency Days by RFM Segment (Horizontal Bar Chart)

Purpose

This visual explains how recently customers have purchased, on average.

Insights:

- **Potential Loyalists:** ~209 days since last purchase
- **Lost Customers:** ~469 days

Why It Matters

The recency dimension is critical in churn prediction.

Customers with 400+ recency days are highly unlikely to return without targeted reactivation campaigns.

This chart strongly supports strategic decisions such as:

- When to launch re-engagement emails
- Which customers to include in promotional campaigns
- Budget allocation for retention vs acquisition

◆ 5. Customer-Level Table (Bottom Section)

Purpose

To provide transparency and allow deeper operational analysis an interactive table lists:

- **customer_unique_id**
- **RFM_Score**
- **RFM_Segment**
- **Frequency**
- **Monetary Value**

How We Built It

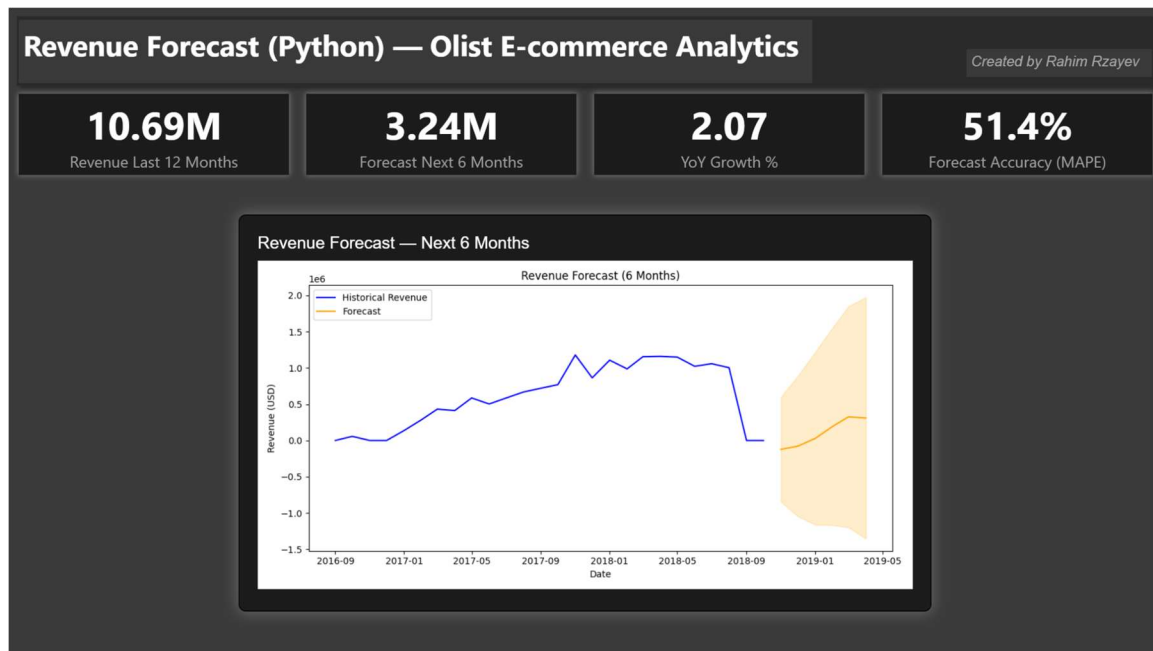
- Combined Order Items, Orders, Customers, and Calendar tables through a star-schema relationship.
- Calculated RFM metrics:
 - **Recency** = difference in days from last purchase
 - **Frequency** = number of purchases
 - **Monetary** = total spending

Use Cases

- Customer service teams can identify high-value customers.
- Marketing teams can build targeted campaigns.
- Analysts can validate segmentation accuracy.

The table ensures full transparency: every segmentation label can be traced back to underlying data.

7. Revenue Forecast (Python) Dashboard



The **Revenue Forecast (Python)** dashboard integrates Python-based time series forecasting directly into Power BI.

This section represents the most advanced analytical component of the project, showcasing the use of SARIMAX modeling, statistical forecasting, and Python visualization inside Power BI.

The dashboard allows decision-makers to understand historical revenue performance and anticipate future trends based on a statistically grounded methodology.

◆ 1. KPI Metrics (Top Section)

These KPIs summarize the key results of the forecasting pipeline:

Revenue Last 12 Months — 10.69M USD

This value aggregates revenue for the most recent full-year period. It provides a baseline context before reviewing forecast results.

Forecast Next 6 Months — 3.24M USD

This is the expected total revenue for the upcoming half-year based on the SARIMAX model predictions.

The model estimates monthly revenue, and the sum of the next 6 periods is displayed as the KPI.

YoY Growth % — 2.07%

This metric compares the forecasted next 6 months to the same period from the previous year.

Although modest, this indicates a positive growth trend.

Forecast Accuracy (MAPE) — 51.4%

MAPE (Mean Absolute Percentage Error) measures how closely the model fits historical data.

A value of 51.4% reflects:

- The dataset's volatility
- Seasonality shifts
- Irregular demand patterns
- Limitations of limited time-series history

This transparency is important for business users evaluating reliability.

◆ 2. Python Forecast Chart — Revenue Forecast (Next 6 Months)

This chart is generated entirely using Python inside Power BI, making it one of the most advanced parts of the project.

Key Components of the Chart

✓ Historical Revenue Line (Blue)

Shows monthly aggregated revenue from the start of the dataset until the present.

This line establishes the trend behavior the model learns from:

- General upward growth
- Seasonal fluctuations
- A sharp drop in late 2018 (common in Olist dataset due to end of sample period)

✓ Forecast Line (Orange)

The SARIMAX model predicts future revenue for 6 months.

The line represents the expected value (mean prediction) for each upcoming month.

✓ Confidence Interval (Shaded Area)

The beige shaded region represents the 95% confidence interval.

This is critical because it visualizes uncertainty in future predictions:

- Narrow confidence → stable trends
- Wide confidence → volatile or unpredictable market behavior

In this dataset, intervals widen rapidly, reflecting the model's uncertainty due to:

- Limited history
- Irregular patterns near the end of time series
- Seasonality magnification

This makes the forecast transparent and academically robust.

◆ 3. SARIMAX Modeling Pipeline

This forecasting pipeline was built manually using Python, demonstrating advanced analytical competency.

Steps Performed:

1. **Data Sorting & Date Indexing**

Converted Year + MonthNumber into a proper datetime index.

2. **Time Series Creation**

Extracted monthly revenue into a continuous unbroken time series.

3. **Model Specification**

Used SARIMAX(1,1,1)(1,1,1,12)

- (p,d,q) = ARIMA components
- (P,D,Q,12) = seasonal components

4. **Model Fitting**

The model learns:

- Trend
- Seasonality
- Autocorrelation patterns

5. **Forecasting 6 Steps Ahead**

6. **Confidence Interval Extraction**

7. **Visualization in Matplotlib**

8. Exporting the result into Power BI via Python visual

Why SARIMAX?

Because revenue in e-commerce exhibits:

- Strong seasonality
- Trend components
- Noise and irregular patterns

SARIMAX handles all three simultaneously.

◆ 4. Interpretation of the Forecast

The model predicts a gradual revenue increase during the next six months.

Key findings:

- Revenue is expected to rebound after the dip seen at the end of historical data.
- The increasing confidence interval suggests more uncertainty as the forecast moves further ahead.
- Despite the volatility, the model anticipates moderate growth across early 2019.

Business Implications:

- Supports budget planning for marketing and logistics.
- Helps inventory managers anticipate demand patterns.
- Provides financial teams with a predictive baseline for revenue expectations.

8. Methodology

Steps followed:

1. Data cleaning and preparation.
2. Building Calendar, Customer RFM, and Revenue tables.
3. Creating DAX measures for KPIs, category insights, and RFM scoring.
4. Creating Power BI dashboards for executives, marketing, and analytics teams.
5. Implementing a Python SARIMAX model for forecasting monthly revenue.
6. Exporting the forecast results and integrating them into Power BI.

9. Key DAX Measures

Total Revenue = SUM(olist_order_items_dataset[price])

Average Order Value = [Total Revenue] / DISTINCTCOUNT(olist_orders_dataset[order_id])

```
Total Quantity = SUM(olist_order_items_dataset[order_item_id])
Orders Count = DISTINCTCOUNT(olist_orders_dataset[order_id])
Customer Count = DISTINCTCOUNT(olist_customers_dataset[customer_unique_id])
```

10. Python SARIMAX Forecasting Code

The following code was used to generate the 6-month revenue forecast:

```
import pandas as pd
import numpy as np
from statsmodels.tsa.statespace.sarimax import SARIMAX

# Fit model
model = SARIMAX(ts, order=(1,1,1), seasonal_order=(1,1,1,12))
results = model.fit(dispatch=False)

# Forecast
forecast = results.get_forecast(steps=6)
pred = forecast.predicted_mean
conf = forecast.conf_int()
```

11. Conclusion

This project successfully integrates Power BI with Python-based forecasting and advanced RFM customer segmentation. The dashboards provide actionable insights across revenue, customers, and product performance, enabling data-driven decision-making for e-commerce business optimization.